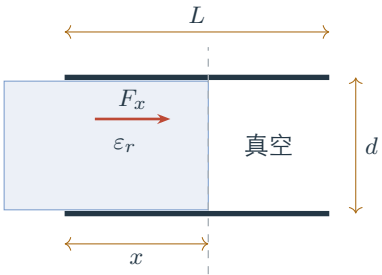
M3-032 部分插入介质板的两种外电路约束

I：接恒压源 V_0；II：断开后 Q_* 固定



纸面外宽 w；$+x$ 沿插入方向向右

I：$F_1 = \frac{1}{2}V_0^2 C'(x)$

断开于 $x = L/2$

$Q_* = C(L/2)V_0$

II：$F_2(x) = \frac{1}{2}[Q_*/C(x)]^2 C'(x)$